

Air Passenger & Route Analysis

1. Abstract

This project presents an end-to-end Airline Passenger & Route Analysis solution using Python, MySQL and Power BI. The analysis integrates flight, passenger, route, airline, airport, aircraft and staff information to evaluate operational performance, route revenue, passenger demand, load factor and customer satisfaction. Python was used for data exploration and cleaning, MySQL was used for structured analysis and aggregation, and Power BI was used to develop an interactive dashboard for decision-making.

2. Introduction

Airline operations generate large volumes of data related to flights, passengers, routes, revenue and service performance. Analysing these data sources together helps identify profitable routes, understand passenger behaviour, monitor aircraft and flight performance, and detect areas that require operational improvement. This project converts raw airline data into cleaned, structured and visual analytical information.

3. Problem Statement

The main problem is to analyse airline operational and passenger data from multiple related datasets and transform it into meaningful business insights. The project focuses on route profitability, passenger demand, load factor, flight status, delays, revenue patterns and passenger satisfaction. The final dashboard should allow users to explore these measures interactively and identify underperforming areas.

4. Objectives

- Clean and prepare airline datasets using Python and Pandas.
- Identify missing values, duplicate records and inconsistent data types.
- Analyse flight, passenger and route information using SQL.
- Measure route-wise revenue, passenger volume and average load factor.
- Evaluate flight status and operational performance.
- Compare airline-level operational performance.
- Study passenger satisfaction across travel classes and flight conditions.

- Build an interactive Power BI dashboard for reporting and decision-making.
- Identify potentially underperforming flights using load factor and delay conditions.

5. Dataset Description

The project uses multiple related datasets:

Dataset	Purpose	Key Information
Flights	Flight and operational analysis	Flight date, airline, route, departure, delay, duration, passengers, revenue, load factor
Passengers	Passenger behaviour and satisfaction	Passenger details, travel class, booking channel, ticket price, satisfaction
Routes	Route-level performance	Origin, destination, flights, passengers, revenue, load factor, average delay
Airlines	Airline reference data	Airline ID, airline name, country
Airports	Airport reference data	Airport ID, airport name, city, country, IATA, coordinates
Aircraft	Aircraft information	Aircraft type, manufacture year, seat capacity
Staffs	Staff analysis	Role, department, experience, salary, base airport, employment status

6. Tools and Technologies

- Python – data loading, exploration, cleaning and analytical preparation.
- Pandas – DataFrame operations, missing-value handling, duplicate removal, grouping and merging.
- NumPy – numerical and conditional operations.
- MySQL – database storage, joins, aggregation and SQL-based analysis.
- Power BI – interactive dashboards, KPIs, charts, slicers, bookmarks and reporting.
- DAX – measures and calculated columns for Power BI analysis.

7. Project Workflow

1. Import raw airline datasets.
2. Explore structure, data types, row counts and summary statistics.
3. Clean missing values and duplicate records using Python.
4. Convert columns to appropriate data types such as date and datetime.
5. Create derived route information and prepare cleaned CSV files.
6. Load the cleaned datasets into MySQL.
7. Perform joins, grouping, aggregation and ranking using SQL.
8. Connect the analytical data to Power BI.
9. Create measures, calculated columns, charts, slicers and navigation bookmarks.
10. Interpret the dashboard results and derive business recommendations.

8. Phase 1 – Python Data Preparation

8.1 Data Import and Exploration

The airline datasets were imported into Python using Pandas. Initial exploration included checking the number of records, column names, data types, summary statistics, missing values and duplicate records. This step helped understand the quality and structure of the raw data.

8.2 Data Cleaning

The cleaning process included removing duplicate records, identifying missing values, correcting data types and filling or validating missing values where appropriate. Date and datetime fields were converted to suitable formats so that month-wise and time-based analysis could be performed reliably.

Examples of data-quality issues handled included missing flight departure information, missing delay values, missing passenger attributes, missing route metrics, missing airport country information and missing aircraft type values.

8.3 Lists, Tuples and Conditional Analysis

Python lists and tuples were used to represent collections such as airline names, airport details and route information. Conditional logic was used to classify flight records based on operational indicators such as delay and load factor.

8.4 Loop-Based Analysis

For-loop and while-loop exercises were used to process flight records, count flights meeting selected conditions and identify highly delayed flights. These exercises provided practical understanding of iterative processing before applying vectorized Pandas operations.

8.5 DataFrame Operations

DataFrames were filtered, sorted, grouped and merged to create analytical datasets. Airport city information was mapped to origin and destination airport IDs to create a readable route name for reporting.

8.6 Statistical and Exploratory Analysis

Statistical analysis was applied to revenue, load factor, passenger satisfaction, flight duration and operational measures. Mean, median, standard deviation and interquartile range were used to understand distributions and identify unusual observations. Correlation analysis and visual exploration were used to study relationships among numerical variables.

```
[2]: import pandas as pd

[3]: flights = pd.read_csv(r"C:\Users\Asus\Downloads\airline_duplicates\Flights.csv")
passengers=pd.read_csv(r"C:\Users\Asus\Downloads\airline_duplicates\Passengers.csv")
routes=pd.read_csv(r"C:\Users\Asus\Downloads\airline_duplicates\Routes.csv")
airlines=pd.read_csv(r"C:\Users\Asus\Downloads\airline_duplicates\Airlines.csv")
airports=pd.read_csv(r"C:\Users\Asus\Downloads\airline_duplicates\Airports.csv")
staffs=pd.read_csv(r"C:\Users\Asus\Downloads\airline_duplicates\Staff.csv")
aircraft=pd.read_csv(r"C:\Users\Asus\Downloads\airline_duplicates\Aircraft.csv")

[4]: flights
```

	Flight_ID	Flight_Date	Airline_ID	Aircraft_ID	Origin_Airport_ID	Destination_Airport_ID	Scheduled_Departure	Actual_Departure	Delay_Minutes	Delay_Reason	
0	FL00001	2023-03-02	AL002	AC013	AP013	AP027	2023-03-02 15:00:00	2023-03-02 15:11:00	11.0	Security	D
1	FL00002	2023-11-18	AL009	AC036	AP010	AP006	2023-11-18 22:30:00	2023-11-18 22:36:00	6.0	Technical	D
2	FL00003	2025-02-05	AL002	AC019	AP022	AP009	2025-02-05 19:30:00	2025-02-05 19:30:00	0.0	NaN	
3	FL00004	2023-11-20	AL004	AC012	AP024	AP030	2023-11-20 13:00:00	2023-11-20 13:12:00	12.0	Late Aircraft	D
4	FL00005	2025-03-30	AL003	AC023	AP023	AP016	2025-03-30 08:15:00	2025-03-30 08:25:00	10.0	Late Aircraft	D
...	...	...	...	...	...	...	...	...	...	...	...
5025	FL04685	2024-04-05	AL008	AC034	AP027	AP012	2024-04-05 21:00:00	2024-04-05 21:14:00	14.0	Late Aircraft	D
5026	FL00290	2024-02-18	AL009	AC010	AP029	AP007	2024-02-18 17:00:00	2024-02-18 17:14:00	14.0	Late Aircraft	D

9. Phase 2 – MySQL Analysis

After cleaning, the datasets were loaded into MySQL for structured analysis. Primary keys and relationships between datasets support joins across flights, airlines, passengers, routes and airports.

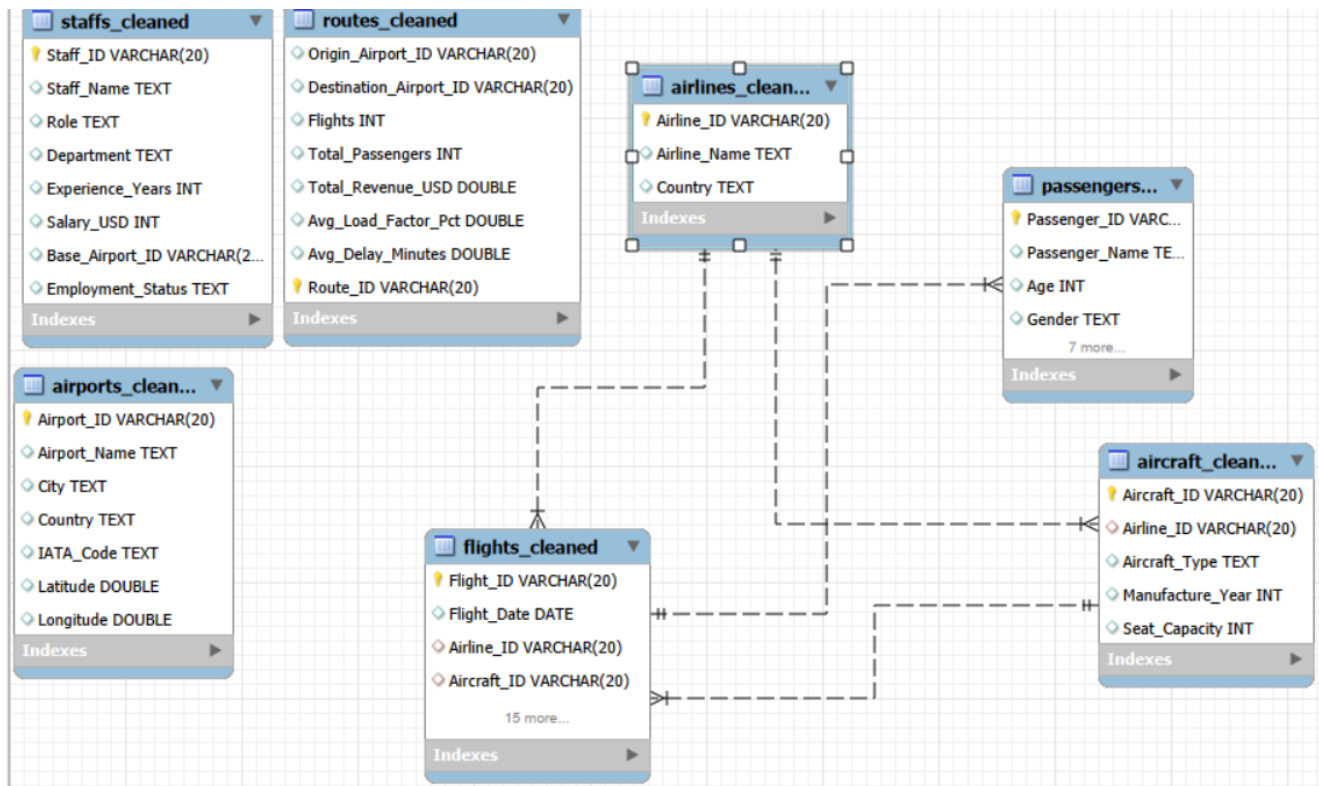
9.1 Major SQL Analysis Areas

- Calculate average delay for each airline and identify the airline with the highest average.

- Calculate route-wise total revenue and rank routes by revenue.
- Calculate route-wise passenger totals and average load factor.
- Identify routes with the highest and lowest average travel time using flight duration.
- Analyse delayed flights and their percentage by airline.
- Compare passenger satisfaction for different flight conditions.
- Aggregate operational measures by season, route and airline.

9.2 Example SQL Query

```
SELECT a.Airline_Name, AVG(f.Delay_Minutes) AS Avg_Delay
FROM airlines_cleaned a
JOIN flights_cleaned f ON a.Airline_ID = f.Airline_ID
GROUP BY a.Airline_Name
ORDER BY Avg_Delay DESC
LIMIT 1;
```



10. Phase 3 – Power BI Dashboard

The cleaned and analysed data was visualized in Power BI. The dashboard was designed to provide a quick overview of revenue, route performance, load factor, flight status and airline operational

performance. Interactive slicers and bookmark-based navigation can be used to organize the analysis into focused sections.

10.1 Key Measures

- Total Revenue = SUM of Revenue_USD
- Total Passengers = SUM of Passengers
- Average Load Factor = AVERAGE of Load_Factor_Pct
- Average Delay = AVERAGE of Delay_Minutes

10.2 Dashboard Visualizations

- Total Revenue by Routes – identifies high-revenue routes.
- Average Load Factor by Month – shows monthly passenger utilization.
- Flight Status Distribution – compares delayed, severely delayed, on-time and cancelled flights.
- Load Factor by Destination City – compares destination-level utilization.
- Revenue by Month – highlights monthly revenue patterns.
- Delay Minutes by Airlines – compares average delay performance among airlines.
- Customer Satisfaction by Travel Class – compares satisfaction across cabin classes.
- Satisfaction Score vs Delay Minutes – explores the relationship between delay and satisfaction.
- Delay Minutes by Season – compares operational performance across seasons.
- Revenue by Season – evaluates seasonal revenue performance.
- Performance Insights – highlights flights with low load factor and high delay.

10.3 Dashboard Preview

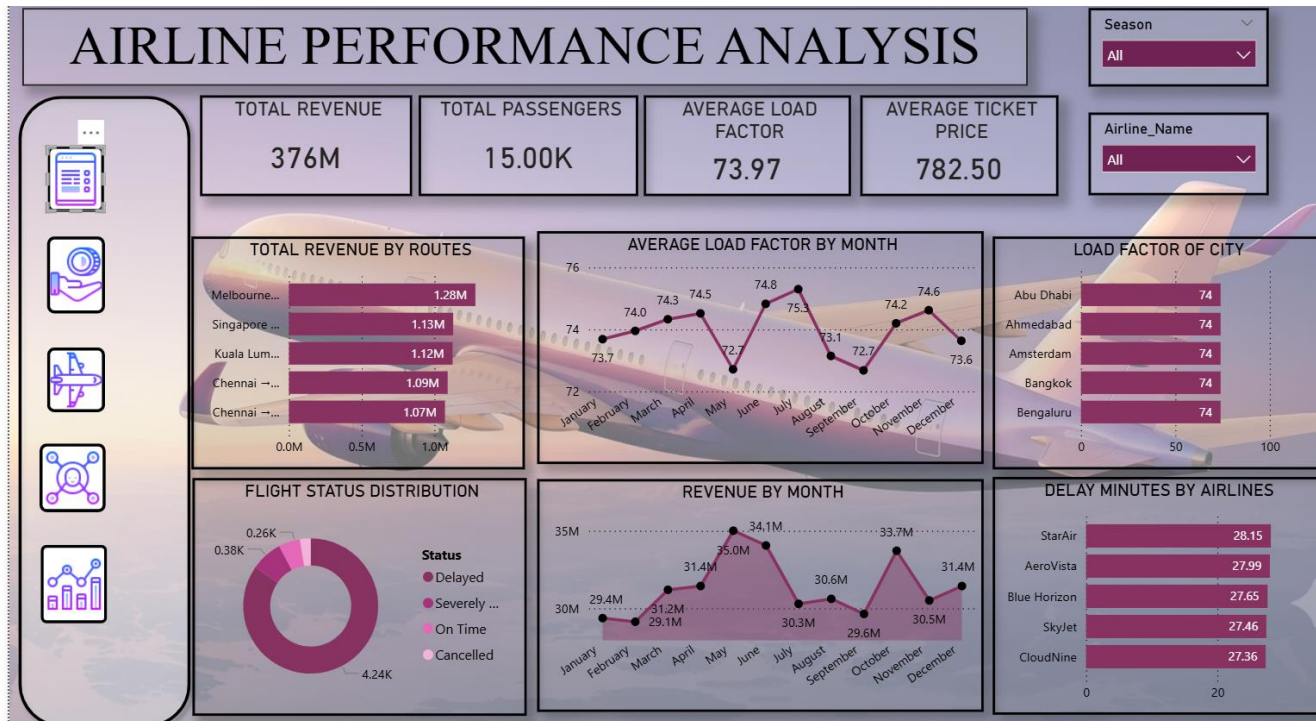


Figure 1: Airline performance dashboard preview

11. Key Findings

- Route revenue varies considerably, allowing high-performing routes to be prioritized for business analysis.
- Monthly load factor shows visible variation across the year, indicating changes in passenger utilization.
- Flight status analysis provides a clear view of the proportion of delayed, severely delayed, on-time and cancelled flights.
- Average delay differs across airlines, making airline-level comparison useful for operational monitoring.
- Monthly revenue changes over time and can be compared with load factor to understand demand and utilization patterns.
- Passenger satisfaction can be analysed together with delay and travel class to identify service-quality patterns.
- Flights with both low load factor and high delay can be treated as potential underperformers for further investigation.

12. Performance Insight Rule

A flight is flagged as potentially underperforming when its load factor is below 50% and its delay is greater than 30 minutes. This combined condition helps focus attention on flights that show both weak passenger utilization and significant operational disruption.

13. Recommendations

- Review low-revenue and low-load routes to determine whether scheduling or capacity adjustments are required.
- Investigate airlines with comparatively high average delays and identify recurring operational causes.
- Use seasonal revenue and load-factor trends to support capacity and scheduling decisions.
- Monitor flights with low load factor and high delay as priority cases for operational review.
- Use passenger satisfaction trends to improve service quality and reduce the customer impact of delays.
- Maintain clean and consistent master data for airports, airlines, routes and aircraft to improve reporting accuracy.

14. Conclusion

The Airline Passenger & Route Analysis project demonstrates a complete data analytics workflow from raw data preparation to business intelligence reporting. Python was used to clean and prepare the datasets, MySQL was used for relational analysis, and Power BI transformed the results into an interactive dashboard. The project provides a practical view of airline revenue, route performance, passenger utilization, flight operations and customer satisfaction. The resulting analysis can support data-driven decisions related to route planning, capacity, operational efficiency and passenger experience.

15. Future Scope

- Develop predictive models for flight delays and passenger demand.
- Forecast route revenue and seasonal passenger volume.
- Add real-time flight tracking data for live operational monitoring.
- Build customer segmentation models using passenger behaviour and satisfaction.
- Automate data refresh and dashboard updates.